Dylan Thornburg | ECEN 511

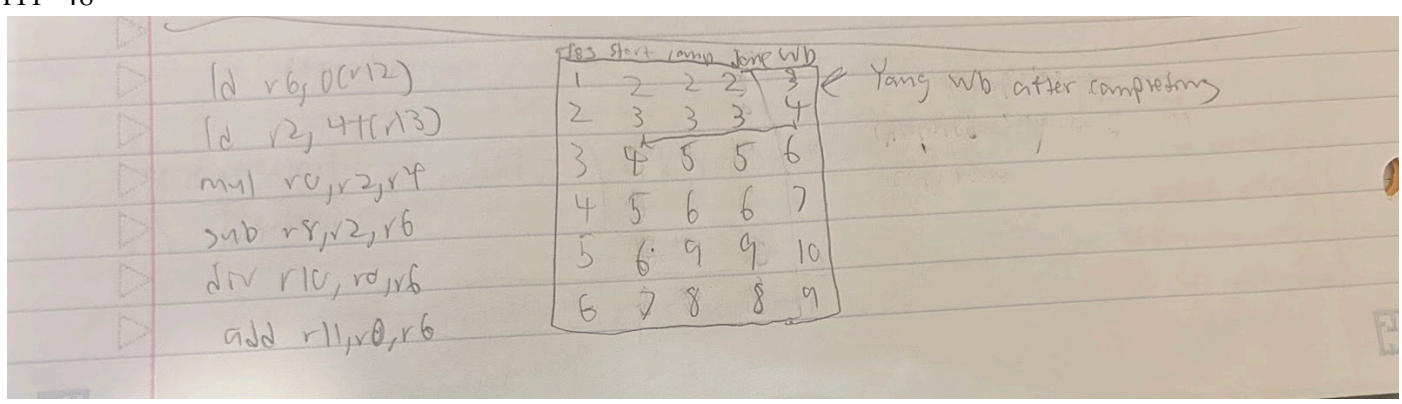
Tomasulo's Algorithm In C

Part 1:

My Tomasulo implementation runs each cycle in the order of write-back, execute, and issue. Only one result can be written back each cycle using the CDB. There is only one mem, one add, and one mul unit. Even though the functional units are freed after WB, a RAW dependent instruction can still start in the same cycle since WB happens before execute. I believe this implementation is what Yang was alluding to when he asks in the slides "why are the instructions in reverse order?"

So for example, for the instrs
Ld r2, 44(r13)
mul r0, r2, r4
When LD writes back, mul will begin execute.
Using the starting sequence, Bufs, and delays we expect 10 clock cycles to complete and regs as follows:

r6=12
r2=9
r0=36
r8=-3
r10=3
r11=48



Delays and FUs:

```
#define NUM_LD_BUF 2
#define NUM_ST_BUF 2
#define NUM_ADD_RS 3
#define NUM_MUL_RS 2

#define LAT_ADD 2 /* executed on ADD */
#define LAT_SUB 2 /* executed on ADD */
#define LAT_MUL 2 /* executed on MUL */
#define LAT_DIV 4 /* executed on MUL */
#define LAT_LD 1 /* executed on Memory Unit */ //chang from 1 -> 2 later
#define LAT_ST 1 /* executed on Memory Unit */
```

And when we run the code we see exactly this behavior expected:

```
============== TEST INSTRUCTION SEQUENCE ===========
I#1    ld     r6,0(r12)
I#2    ld     r2,44(r13)
I#3    mul    r0,r2,r4
I#4    sub    r8,r2,r6
I#5    div    r10,r0,r6
I#6    add    r11,r0,r6
* CYCLE 0 (initial state)
==========================================================================================
===================
RS_id  type   Busy   inst#  Op    Vj    Vk    Qj    Qj    A     Exec  Done
------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#2     LD     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#3     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#4     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#5     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#6     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#7     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#8     MUL    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#9     MUL    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
==========================================================================================
===================
==========================================================================
Registers
--------------------------------------------------------------------
      r0    r1    r2    r3    r4    r5    r6    r7
--------------------------------------------------------------------
val   0     1     2     3     4     5     6     7
Qi    #0    #0    #0    #0    #0    #0    #0    #0
--------------------------------------------------------------------
      r8    r9    r10   r11   r12   r13   r14   r15
--------------------------------------------------------------------
val   8     9     10    11    12    13    14    15
Qi    #0    #0    #0    #0    #0    #0    #0    #0
--------------------------------------------------------------------
==========================================================================
* CYCLE 1
==========================================================================================
===================
RS_id  type   Busy   inst#  Op    Vj    Vk    Qj    Qj    A     Exec  Done
------------------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld    12    -1    #0    #0    0     No    No
#2     LD     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#3     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#4     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#5     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#6     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#7     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#8     MUL    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
```

```
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==========================================================================================
==================
=================================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      2      3      4      5      6      7
Qi     #0     #0     #0     #0     #0     #0     #1     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================================
* CYCLE 2
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld     12     -1     #0     #0     12     Yes    Yes
#2     LD     Yes    I#-1   ld     13     -1     #0     #0     44     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==========================================================================================
==================
=================================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      2      3      4      5      6      7
Qi     #0     #0     #2     #0     #0     #0     #1     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================================
* CYCLE 3
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     Yes    I#-1   ld     13     -1     #0     #0     57     Yes    Yes
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
```

```
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    -1     4      #2     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      2      3      4      5      12     7
Qi     #8     #0     #2     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 4
==================================================================================
==================
RS_id  type   Busy   inst#  Op     Vj     Vk     Qj     Qj     A      Exec   Done
----------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    Yes    I#-1   sub    9      12     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    9      4      #0     #0     -1     Yes    No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      9      3      4      5      12     7
Qi     #8     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #5     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 5
==================================================================================
==================
```

```
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    Yes    I#-1   sub    9      12     #0     #0     -1     Yes    No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    9      4      #0     #0     -1     No     Yes
#9     MUL    Yes    I#-1   div    -1     12     #8     #0     -1     No     No
=============================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      9      3      4      5      12     7
Qi     #8     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #5     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 6
=============================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    Yes    I#-1   sub    9      12     #0     #0     -1     No     Yes
#6     ADD    Yes    I#-1   add    36     12     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    Yes    I#-1   div    36     12     #0     #0     -1     Yes    No
=============================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    36     1      9      3      4      5      12     7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #5     #0     #9     #6     #0     #0     #0     #0
```

```
-----------------------------------------------------------------
=========================================================================
* CYCLE 7
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
---------------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    Yes    I#-1   add    36     12     #0     #0     -1     Yes    No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    Yes    I#-1   div    36     12     #0     #0     -1     Yes    No
=================================================================================
==================
=========================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    36     1      9      3      4      5      12     7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    -3     9      10     11     12     13     14     15
Qi     #0     #0     #9     #6     #0     #0     #0     #0
-----------------------------------------------------------------
=========================================================================
* CYCLE 8
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
---------------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    Yes    I#-1   add    36     12     #0     #0     -1     No     Yes
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    Yes    I#-1   div    36     12     #0     #0     -1     Yes    No
=================================================================================
==================
=========================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    36     1      9      3      4      5      12     7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
```

```
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    -3     9      10     11     12     13     14     15
Qi     #0     #0     #9     #6     #0     #0     #0     #0
-----------------------------------------------------------------
==========================================================================
* CYCLE 9
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
---------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    Yes    I#-1   div    36     12     #0     #0     -1     No     Yes
=================================================================================
==================
==========================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    36     1      9      3      4      5      12     7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    -3     9      10     48     12     13     14     15
Qi     #0     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------
==========================================================================
* CYCLE 10
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
---------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
=================================================================================
==================
==========================================================================
Registers
```

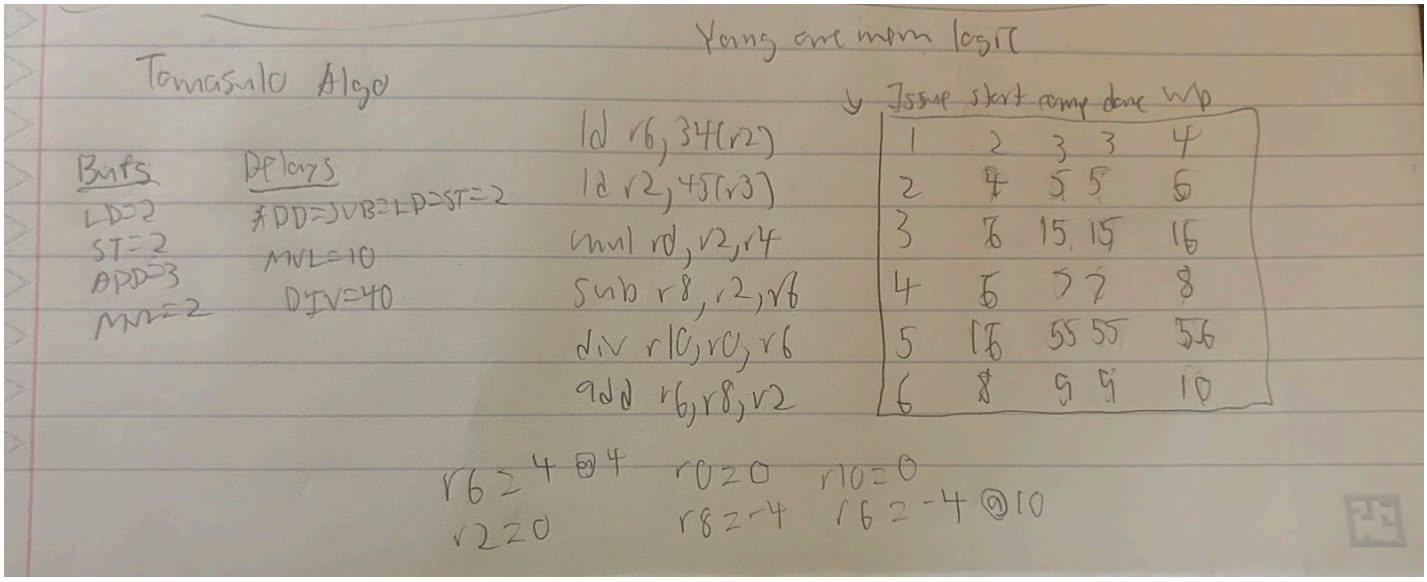
```
-----------------------------------------------------------------
      r0    r1    r2    r3    r4    r5    r6    r7
-----------------------------------------------------------------
val   36    1     9     3     4     5     12    7
Qi    #0    #0    #0    #0    #0    #0    #0    #0
-----------------------------------------------------------------
      r8    r9    r10   r11   r12   r13   r14   r15
-----------------------------------------------------------------
val   -3    9     3     48    12    13    14    15
Qi    #0    #0    #0    #0    #0    #0    #0    #0
-----------------------------------------------------------------
=================================================================================
```

The reservation station table and register values are exactly what we expect.

When LD is changed to delay 2, we see it take 13 cycles which is what was expected

```
==========================================================================
* CYCLE 13
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A     Exec   Done
-----------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==========================================================================================
==================
==========================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    36     1      9      3      4      5      12     7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    -3     9      3      48     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
==========================================================================
```

Let’s take a look at another example




```
============== TEST INSTRUCTION SEQUENCE ===========
I#1    ld     r6,34(r2)
I#2    ld     r2,45(r3)
I#3    mul    r0,r2,r4
I#4    sub    r8,r2,r6
I#5    div    r10,r0,r6
I#6    add    r6,r8,r2
* CYCLE 0 (initial state)
==================================================================================================
===================
RS_id  type   Busy   inst#  Op    Vj    Vk    Qj    Qj    A     Exec  Done
-----------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#2     LD     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#3     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#4     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#5     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#6     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#7     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#8     MUL    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#9     MUL    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
==================================================================================================
===================
=========================================================================
Registers
```

```
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      2      3      4      5      6      7
Qi    #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi    #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 1
=================================================================================
===================
RS_id  type   Busy   inst#  Op     Vj     Vk     Qj     Qj     A      Exec   Done
---------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld     2      -1     #0     #0     34     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
=================================================================================
===================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      2      3      4      5      6      7
Qi    #0     #0     #0     #0     #0     #0     #1     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi    #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 2
=================================================================================
===================
RS_id  type   Busy   inst#  Op     Vj     Vk     Qj     Qj     A      Exec   Done
---------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld     2      -1     #0     #0     36     Yes    No
#2     LD     Yes    I#-1   ld     3      -1     #0     #0     45     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
```

```
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==========================================================================================
==================
=================================================================================
Registers
-----------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------------
val    0      1      2      3      4      5      6      7
Qi     #0     #0     #2     #0     #0     #0     #1     #0
-----------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
=================================================================================
* CYCLE 3
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld     2      -1     #0     #0     36     No     Yes
#2     LD     Yes    I#-1   ld     3      -1     #0     #0     45     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    -1     4      #2     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==========================================================================================
==================
=================================================================================
Registers
-----------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------------
val    0      1      2      3      4      5      6      7
Qi     #8     #0     #2     #0     #0     #0     #1     #0
-----------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
=================================================================================
* CYCLE 4
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     Yes    I#-1   ld     3      -1     #0     #0     48     Yes    No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
```

```
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    Yes    I#-1   sub    -1     4      #2     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    -1     4      #2     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==============================================================================================
===================
=========================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      2      3      4      5      4      7
Qi     #8     #0     #2     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #5     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=========================================================================
* CYCLE 5
==============================================================================================
===================
RS_id  type   Busy   inst#  Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     Yes    I#-1   ld     3      -1     #0     #0     48     No     Yes
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    Yes    I#-1   sub    -1     4      #2     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    -1     4      #2     #0     -1     No     No
#9     MUL    Yes    I#-1   div    -1     4      #8     #0     -1     No     No
==============================================================================================
===================
=========================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      2      3      4      5      4      7
Qi     #8     #0     #2     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #5     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=========================================================================
* CYCLE 6
==============================================================================================
===================
```

```
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    Yes    I#-1   sub    0      4      #0     #0     -1     Yes    No
#6     ADD    Yes    I#-1   add    -1     0      #5     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    0      4      #0     #0     -1     Yes    No
#9     MUL    Yes    I#-1   div    -1     4      #8     #0     -1     No     No
=============================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      0      3      4      5      4      7
Qi     #8     #0     #0     #0     #0     #0     #6     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #5     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 7
=============================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    Yes    I#-1   sub    0      4      #0     #0     -1     No     Yes
#6     ADD    Yes    I#-1   add    -1     0      #5     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    0      4      #0     #0     -1     Yes    No
#9     MUL    Yes    I#-1   div    -1     4      #8     #0     -1     No     No
=============================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      0      3      4      5      4      7
Qi     #8     #0     #0     #0     #0     #0     #6     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #5     #0     #9     #0     #0     #0     #0     #0
```

```
-----------------------------------------------------------------
=========================================================================
* CYCLE 8
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
---------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    Yes    I#-1   add    -4     0      #0     #0     -1     Yes    No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    0      4      #0     #0     -1     Yes    No
#9     MUL    Yes    I#-1   div    -1     4      #8     #0     -1     No     No
=================================================================================
==================
=========================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      0      3      4      5      4      7
Qi     #8     #0     #0     #0     #0     #0     #6     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    -4     9      10     11     12     13     14     15
Qi     #0     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=========================================================================
* CYCLE 9
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
---------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    Yes    I#-1   add    -4     0      #0     #0     -1     No     Yes
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    0      4      #0     #0     -1     Yes    No
#9     MUL    Yes    I#-1   div    -1     4      #8     #0     -1     No     No
=================================================================================
==================
=========================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      0      3      4      5      4      7
Qi     #8     #0     #0     #0     #0     #0     #6     #0
```

```
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    -4     9      10     11     12     13     14     15
Qi     #0     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------
==========================================================================
* CYCLE 10
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
---------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    0      4      #0     #0     -1     Yes    No
#9     MUL    Yes    I#-1   div    -1     4      #8     #0     -1     No     No
=================================================================================
==================
==========================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      0      3      4      5      -4     7
Qi     #8     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    -4     9      10     11     12     13     14     15
Qi     #0     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------
==========================================================================
* CYCLE 11
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
---------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    0      4      #0     #0     -1     Yes    No
#9     MUL    Yes    I#-1   div    -1     4      #8     #0     -1     No     No
=================================================================================
==================
==========================================================================
Registers
```

```
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      0      3      4      5     -4      7
Qi    #8     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9    r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val   -4      9     10     11     12     13     14     15
Qi    #0     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 12
=================================================================================
===================
RS_id  type   Busy   inst#  Op    Vj    Vk    Qj    Qj    A     Exec   Done
---------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#2     LD     No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#3     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#4     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#5     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#6     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#7     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#8     MUL    Yes    I#-1   mul   0     4     #0    #0    -1    Yes    No
#9     MUL    Yes    I#-1   div   -1    4     #8    #0    -1    No     No
=================================================================================
===================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      0      3      4      5     -4      7
Qi    #8     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9    r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val   -4      9     10     11     12     13     14     15
Qi    #0     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 13
=================================================================================
===================
RS_id  type   Busy   inst#  Op    Vj    Vk    Qj    Qj    A     Exec   Done
---------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#2     LD     No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#3     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#4     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#5     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#6     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#7     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No     No
#8     MUL    Yes    I#-1   mul   0     4     #0    #0    -1    Yes    No
```

```
#9     MUL    Yes    I#-1   div    -1     4      #8     #0     -1     No     No
==========================================================================================
==================
=================================================================================
Registers
-----------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------------
val    0      1      0      3      4      5     -4      7
Qi     #8     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------------
val   -4      9     10     11     12     13     14     15
Qi     #0     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
=================================================================================
* CYCLE 14
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    0      4      #0     #0     -1     Yes    No
#9     MUL    Yes    I#-1   div    -1     4      #8     #0     -1     No     No
==========================================================================================
==================
=================================================================================
Registers
-----------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------------
val    0      1      0      3      4      5     -4      7
Qi     #8     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------------
val   -4      9     10     11     12     13     14     15
Qi     #0     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
=================================================================================
* CYCLE 15
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
```

```
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    0      4      #0     #0     -1     No     Yes
#9     MUL    Yes    I#-1   div    -1     4      #8     #0     -1     No     No
==================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      0      3      4      5      -4     7
Qi     #8     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    -4     9      10     11     12     13     14     15
Qi     #0     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 16
==================================================================================
==================
RS_id  type   Busy   inst#  Op     Vj     Vk     Qj     Qj     A      Exec   Done
----------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    Yes    I#-1   div    0      4      #0     #0     -1     Yes    No
==================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      0      3      4      5      -4     7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    -4     9      10     11     12     13     14     15
Qi     #0     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
```

… SKIP TO CYCLE 55 …

```
* CYCLE 55
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    Yes    I#-1   div    0      4      #0     #0     -1     No     Yes
==========================================================================================
==================
==========================================================================
Registers
-----------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------------
val    0      1      0      3      4      5     -4      7
Qi    #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------------
val   -4      9     10     11     12     13     14     15
Qi    #0     #0     #9     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
==========================================================================
* CYCLE 56
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==========================================================================================
==================
==========================================================================
Registers
-----------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------------
val    0      1      0      3      4      5     -4      7
Qi    #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
```

```
-----------------------------------------------------------------
val    -4     9     0     11     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
==============================================================================
```

This result also matches exactly what we expect.

Last Example, just change the first instruction to ST so we test ALL instruction types. Everything except for the start is the same for the reservation stations, however the final register values change. Also it still takes 56 cycles because ST delay = LD delay = 2.
Expected:
r2 = 0
r0 = 0
r8 = -6
r10 = 0
r6 = -6

Note: the first instruction kept saying ld until I fixed an error in Yang's inst.c code.

```
============== TEST INSTRUCTION SEQUENCE ===========
I#1    st     r6,34(r2)
I#2    ld     r2,45(r3)
I#3    mul    r0,r2,r4
I#4    sub    r8,r2,r6
I#5    div    r10,r0,r6
I#6    add    r6,r8,r2
* CYCLE 0 (initial state)
==========================================================================================
==================
RS_id  type  Busy  inst#  Op    Vj    Vk    Qj    Qj    A     Exec  Done
------------------------------------------------------------------------------------------
#1     LD    No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#2     LD    No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#3     ST    No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#4     ST    No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#5     ADD   No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#6     ADD   No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#7     ADD   No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#8     MUL   No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#9     MUL   No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
==========================================================================================
==================
==============================================================================
Registers
------------------------------------------------------------------
      r0    r1    r2    r3    r4    r5    r6    r7
------------------------------------------------------------------
val   0     1     2     3     4     5     6     7
Qi    #0    #0    #0    #0    #0    #0    #0    #0
------------------------------------------------------------------
      r8    r9    r10   r11   r12   r13   r14   r15
------------------------------------------------------------------
val   8     9     10    11    12    13    14    15
Qi    #0    #0    #0    #0    #0    #0    #0    #0
------------------------------------------------------------------
==============================================================================
* CYCLE 1
```

```
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A     Exec   Done
------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     Yes    I#-1   st     2      6      #0     #0     34     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==========================================================================================
==================
=========================================================================
Registers
-------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-------------------------------------------------------------------
val    0      1      2      3      4      5      6      7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-------------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-------------------------------------------------------------------
=========================================================================
* CYCLE 2
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A     Exec   Done
------------------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld     3      -1     #0     #0     45     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     Yes    I#-1   st     2      6      #0     #0     36     Yes    No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==========================================================================================
==================
=========================================================================
Registers
-------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-------------------------------------------------------------------
val    0      1      2      3      4      5      6      7
Qi     #0     #0     #1     #0     #0     #0     #0     #0
-------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-------------------------------------------------------------------
```

```
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 3
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld     3      -1     #0     #0     45     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     Yes    I#-1   st     2      6      #0     #0     36     No     Yes
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    -1     4      #1     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
=================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      2      3      4      5      6      7
Qi     #8     #0     #1     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 4
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld     3      -1     #0     #0     48     Yes    No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    Yes    I#-1   sub    -1     6      #1     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    -1     4      #1     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
=================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
```

```
val    0     1     2     3     4     5     6     7
Qi    #8    #0    #1    #0    #0    #0    #0    #0
-----------------------------------------------------------------
      r8    r9    r10   r11   r12   r13   r14   r15
-----------------------------------------------------------------
val    8     9     10    11    12    13    14    15
Qi    #5    #0    #0    #0    #0    #0    #0    #0
-----------------------------------------------------------------
```

… SKIP TO CYCLE 56…

```
=================================================================
* CYCLE 56
=================================================================================================
==================
RS_id  type  Busy  inst#  Op    Vj    Vk    Qj    Qj    A     Exec  Done
-----------------------------------------------------------------------------------------------
#1     LD    No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#2     LD    No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#3     ST    No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#4     ST    No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#5     ADD   No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#6     ADD   No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#7     ADD   No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#8     MUL   No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
#9     MUL   No    I#-1   NONE  -1    -1    #0    #0    -1    No    No
=================================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0    r1    r2    r3    r4    r5    r6    r7
-----------------------------------------------------------------
val    0     1     0     3     4     5    -6     7
Qi    #0    #0    #0    #0    #0    #0    #0    #0
-----------------------------------------------------------------
      r8    r9    r10   r11   r12   r13   r14   r15
-----------------------------------------------------------------
val   -6     9     0     11    12    13    14    15
Qi    #0    #0    #0    #0    #0    #0    #0    #0
-----------------------------------------------------------------
=================================================================
```

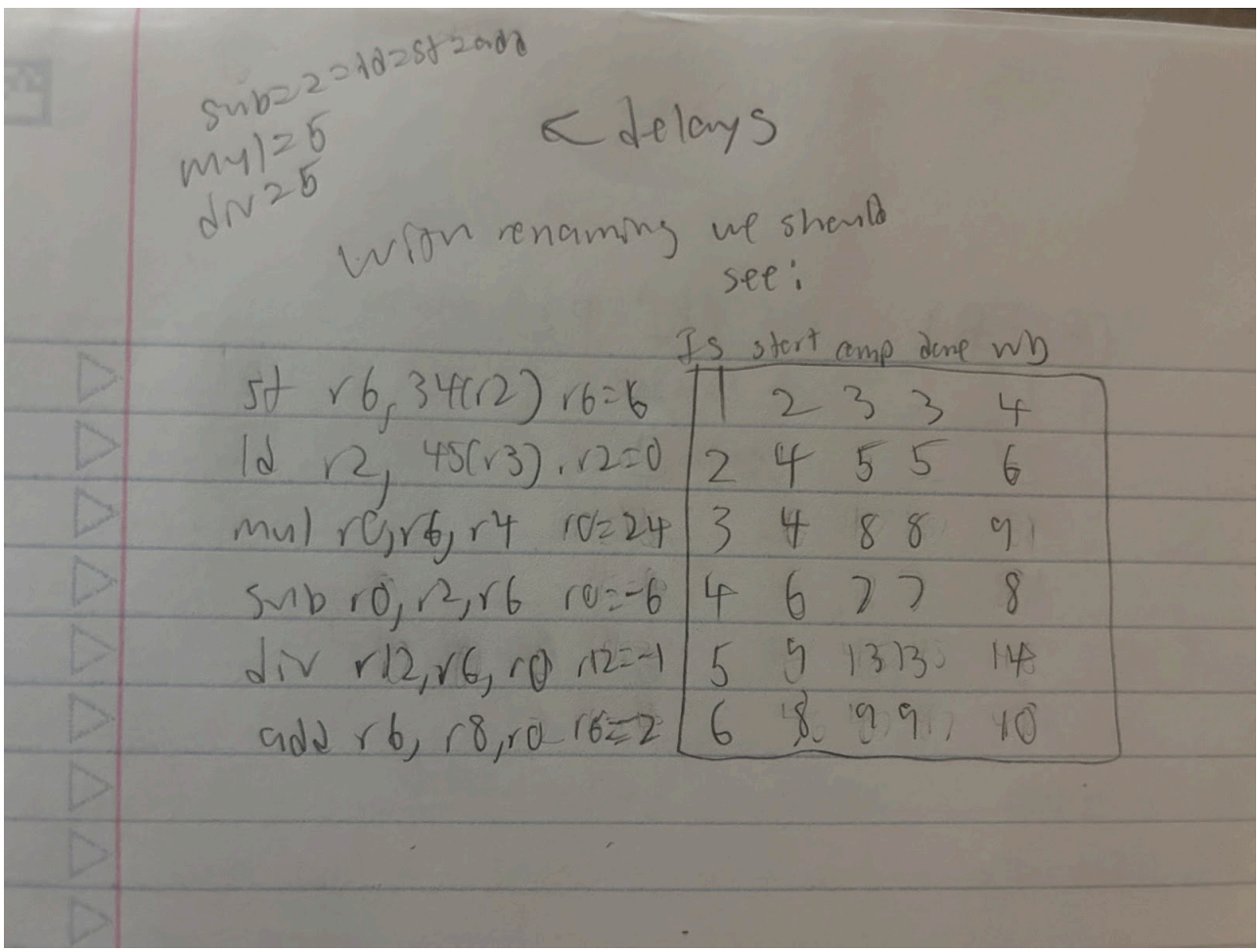
As we can see, the results are different now as we stored r6 to 34(r2) instead of loading from 34(r2).

PART 2: Renaming

Right off the bat, I don't actually build a separate physical register file for the renaming. Instead I use a RS tag. I simply tag the RS IDs to track which instruction produces each value, so my code behaves as if it had renaming by default. The CDB wakes dependents by tag, and the regfile only commits if the tag still matches. WAW/WAR hazards essentially vanish if the tag doesn't match even without a physical register file.

Delays =2 except mult=div=5.

EXAMPLE:




```
============== TEST INSTRUCTION SEQUENCE ===========
I#1    st     r6,34(r2)
I#2    ld     r2,45(r3)
```

```
I#3     mul     r0,r6,r4
I#4     sub     r0,r2,r6
I#5     div     r12,r6,r0
I#6     add     r6,r8,r0
* CYCLE 0 (initial state)
==================================================================================
===================
RS_id   type    Busy    inst#   Op      Vj      Vk      Qj      Qj      A       Exec    Done
----------------------------------------------------------------------------------------------
#1      LD      No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#2      LD      No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#3      ST      No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#4      ST      No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#5      ADD     No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#6      ADD     No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#7      ADD     No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#8      MUL     No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#9      MUL     No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
==================================================================================
===================
=========================================================================
Registers
-----------------------------------------------------------------
      r0      r1      r2      r3      r4      r5      r6      r7
-----------------------------------------------------------------
val     0       1       2       3       4       5       6       7
Qi      #0      #0      #0      #0      #0      #0      #0      #0
-----------------------------------------------------------------
      r8      r9      r10     r11     r12     r13     r14     r15
-----------------------------------------------------------------
val     8       9       10      11      12      13      14      15
Qi      #0      #0      #0      #0      #0      #0      #0      #0
-----------------------------------------------------------------
=========================================================================
* CYCLE 1
==================================================================================
===================
RS_id   type    Busy    inst#   Op      Vj      Vk      Qj      Qj      A       Exec    Done
----------------------------------------------------------------------------------------------
#1      LD      No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#2      LD      No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#3      ST      Yes     I#-1    st      2       6       #0      #0      34      No      No
#4      ST      No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#5      ADD     No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#6      ADD     No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#7      ADD     No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#8      MUL     No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
#9      MUL     No      I#-1    NONE    -1      -1      #0      #0      -1      No      No
==================================================================================
===================
=========================================================================
Registers
-----------------------------------------------------------------
      r0      r1      r2      r3      r4      r5      r6      r7
-----------------------------------------------------------------
```

```
val    0     1     2     3     4     5     6     7
Qi    #0    #0    #0    #0    #0    #0    #0    #0
-----------------------------------------------------------------------
      r8    r9    r10   r11   r12   r13   r14   r15
-----------------------------------------------------------------------
val    8     9     10    11    12    13    14    15
Qi    #0    #0    #0    #0    #0    #0    #0    #0
-----------------------------------------------------------------------
=========================================================================
* CYCLE 2
=================================================================================
===================
RS_id  type   Busy   inst#  Op    Vj    Vk    Qj    Qj    A     Exec  Done
---------------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld    3     -1    #0    #0    45    No    No
#2     LD     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#3     ST     Yes    I#-1   st    2     6     #0    #0    36    Yes   No
#4     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#5     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#6     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#7     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#8     MUL    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#9     MUL    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
=================================================================================
==================
=========================================================================
Registers
-----------------------------------------------------------------------
      r0    r1    r2    r3    r4    r5    r6    r7
-----------------------------------------------------------------------
val    0     1     2     3     4     5     6     7
Qi    #0    #0    #1    #0    #0    #0    #0    #0
-----------------------------------------------------------------------
      r8    r9    r10   r11   r12   r13   r14   r15
-----------------------------------------------------------------------
val    8     9     10    11    12    13    14    15
Qi    #0    #0    #0    #0    #0    #0    #0    #0
-----------------------------------------------------------------------
=========================================================================
* CYCLE 3
=================================================================================
===================
RS_id  type   Busy   inst#  Op    Vj    Vk    Qj    Qj    A     Exec  Done
---------------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld    3     -1    #0    #0    45    No    No
#2     LD     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#3     ST     Yes    I#-1   st    2     6     #0    #0    36    No    Yes
#4     ST     No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#5     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#6     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#7     ADD    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
#8     MUL    Yes    I#-1   mul   6     4     #0    #0    -1    No    No
#9     MUL    No     I#-1   NONE  -1    -1    #0    #0    -1    No    No
=================================================================================
==================
```

```
==========================================================================
Registers
--------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
--------------------------------------------------------------------
val    0      1      2      3      4      5      6      7
Qi     #8     #0     #1     #0     #0     #0     #0     #0
--------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
--------------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
--------------------------------------------------------------------
==========================================================================
* CYCLE 4
==============================================================================================
====================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
----------------------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld     3      -1     #0     #0     48     Yes    No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    Yes    I#-1   sub    -1     6      #1     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    6      4      #0     #0     -1     Yes    No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
==============================================================================================
===================
==========================================================================
Registers
--------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
--------------------------------------------------------------------
val    0      1      2      3      4      5      6      7
Qi     #5     #0     #1     #0     #0     #0     #0     #0
--------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
--------------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
--------------------------------------------------------------------
==========================================================================
* CYCLE 5
==============================================================================================
====================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
----------------------------------------------------------------------------------------------
#1     LD     Yes    I#-1   ld     3      -1     #0     #0     48     No     Yes
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    Yes    I#-1   sub    -1     6      #1     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
```

```
#7    ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8    MUL    Yes    I#-1   mul    6      4      #0     #0     -1     Yes    No
#9    MUL    Yes    I#-1   div    6      -1     #0     #5     -1     No     No
==================================================================================
===================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val   0      1      2      3      4      5      6      7
Qi    #5     #0     #1     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val   8      9      10     11     12     13     14     15
Qi    #0     #0     #0     #0     #9     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 6
==================================================================================
===================
RS_id  type   Busy   inst#  Op     Vj     Vk     Qj     Qj     A      Exec   Done
----------------------------------------------------------------------------------
#1    LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2    LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3    ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4    ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5    ADD    Yes    I#-1   sub    0      6      #0     #0     -1     Yes    No
#6    ADD    Yes    I#-1   add    8      -1     #0     #5     -1     No     No
#7    ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8    MUL    Yes    I#-1   mul    6      4      #0     #0     -1     Yes    No
#9    MUL    Yes    I#-1   div    6      -1     #0     #5     -1     No     No
==================================================================================
===================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val   0      1      0      3      4      5      6      7
Qi    #5     #0     #0     #0     #0     #0     #6     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val   8      9      10     11     12     13     14     15
Qi    #0     #0     #0     #0     #9     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 7
==================================================================================
===================
RS_id  type   Busy   inst#  Op     Vj     Vk     Qj     Qj     A      Exec   Done
----------------------------------------------------------------------------------
#1    LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
```

```
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    Yes    I#-1   sub    0      6      #0     #0     -1     No     Yes
#6     ADD    Yes    I#-1   add    8      -1     #0     #5     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    6      4      #0     #0     -1     Yes    No
#9     MUL    Yes    I#-1   div    6      -1     #0     #5     -1     No     No
==================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    0      1      0      3      4      5      6      7
Qi     #5     #0     #0     #0     #0     #0     #6     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #9     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 8
==================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    Yes    I#-1   add    8      -6     #0     #0     -1     Yes    No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    Yes    I#-1   mul    6      4      #0     #0     -1     No     Yes
#9     MUL    Yes    I#-1   div    6      -6     #0     #0     -1     No     No
==================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    -6     1      0      3      4      5      6      7
Qi     #0     #0     #0     #0     #0     #0     #6     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #9     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 9
```

```
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A     Exec   Done
------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    Yes    I#-1   add    8      -6     #0     #0     -1     No     Yes
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    Yes    I#-1   div    6      -6     #0     #0     -1     Yes    No
==========================================================================================
==================
==========================================================================
Registers
--------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
--------------------------------------------------------------------
val    -6     1      0      3      4      5      6      7
Qi     #0     #0     #0     #0     #0     #0     #6     #0
--------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
--------------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #9     #0     #0     #0
--------------------------------------------------------------------
==========================================================================
* CYCLE 10
==========================================================================================
==================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A     Exec   Done
------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    Yes    I#-1   div    6      -6     #0     #0     -1     Yes    No
==========================================================================================
==================
==========================================================================
Registers
--------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
--------------------------------------------------------------------
val    -6     1      0      3      4      5      2      7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
--------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
--------------------------------------------------------------------
```

```
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #9     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 11
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    Yes    I#-1   div    6      -6     #0     #0     -1     Yes    No
=================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
val    -6     1      0      3      4      5      2      7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #9     #0     #0     #0
-----------------------------------------------------------------
=================================================================
* CYCLE 12
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    Yes    I#-1   div    6      -6     #0     #0     -1     Yes    No
=================================================================================
==================
=================================================================
Registers
-----------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------
```

```
val    -6     1      0      3      4      5      2      7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
       r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #9     #0     #0     #0
-----------------------------------------------------------------------
=========================================================================
* CYCLE 13
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    Yes    I#-1   div    6      -6     #0     #0     -1     No     Yes
=================================================================================
==================
=========================================================================
Registers
-----------------------------------------------------------------------
       r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------------
val    -6     1      0      3      4      5      2      7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
       r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------------
val    8      9      10     11     12     13     14     15
Qi     #0     #0     #0     #0     #9     #0     #0     #0
-----------------------------------------------------------------------
=========================================================================
* CYCLE 14
=================================================================================
===================
RS_id   type   Busy   inst#   Op     Vj     Vk     Qj     Qj     A      Exec   Done
-----------------------------------------------------------------------------------------------
#1     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#2     LD     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#3     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#4     ST     No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#5     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#6     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#7     ADD    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#8     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
#9     MUL    No     I#-1   NONE   -1     -1     #0     #0     -1     No     No
=================================================================================
==================
```

```
==============================================================================
Registers
-----------------------------------------------------------------------
      r0     r1     r2     r3     r4     r5     r6     r7
-----------------------------------------------------------------------
val    -6     1      0      3      4      5      2      7
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
      r8     r9     r10    r11    r12    r13    r14    r15
-----------------------------------------------------------------------
val    8      9      10     11     -1     13     14     15
Qi     #0     #0     #0     #0     #0     #0     #0     #0
-----------------------------------------------------------------------
==============================================================================
```

You'll see that mul r0 never gets wb to r0. My renaming method effectively avoids these WAR and WAW hazards without ever actually having to create renamed data structures.